

Vivek Sharma

Data Engineering Lead | Data Architect | Engineering Leader

Email: vivek051012@gmail.com | Phone: +91-9762134796

PROFESSIONAL SUMMARY

Data Engineering Lead at **CertifyOS**, leading the design and delivery of large-scale healthcare compliance and provider attribution systems. I architect pipelines, mentor engineers, and align across cross-functional teams. With deep expertise in the modern data stack — **Apache Airflow**, **dbt**, **Airbyte**, and **Snowflake** — alongside GCP, BigQuery, and API integration, I've built reusable, automated ELT systems that power critical workflows, from NPPES integrations to NCQA license compliance and sanctions ingestion. Now seeking to grow further into an **Engineering Manager** or **Data Architect** role where I can scale systems, people, and outcomes.

CORE COMPETENCIES

- **Modern Data Stack:** Apache Airflow, dbt, Airbyte, Snowflake
 - **Data Architecture & Modeling:** SCD Type 2, dbt models (staging, marts, snapshots), temporal tracking, config-driven ingestion
 - **Cloud Data Warehouses:** Snowflake, BigQuery, Amazon Redshift, Singlestore
 - **GCP Stack:** BigQuery, GCS, Pub/Sub, Cloud Composer (Airflow), Dataflow
 - **Languages & Tools:** Python, SQL, Bash, GitHub Actions, Docker
 - **API & Schema Integration:** Nested JSON mapping, CertifyOS DAL, NPPES, license payloads
 - **ETL & Orchestration:** Airflow (Cloud Composer), Airbyte, Pub/Sub triggers, artifact versioning
 - **Leadership:** Sprint planning, stakeholder communication, mentoring
 - **Cross-Functional Collaboration:** Platform, Pipeline, Scraper, and Data Science teams
-

TEAM & LEADERSHIP INVOLVEMENT

- Present architectural designs to **Engineering Managers, Directors**, and **Chief Architects**
 - Participate in **sprint planning**, story grooming, and defining deliverables with PMs
 - Lead technical decision-making across Data, Platform, and Scraper teams
 - Mentor junior engineers and analysts on dbt modeling, Airbyte ingestion, BigQuery, and automation
 - Drive best practices in CI/CD, schema evolution, and DAG observability
-

ONGOING & RECENT PROJECTS

Project	Role	Description
Snowflake Compliance Data Marts	Architect & Lead	Designed dbt models on Snowflake serving provider-compliance and license-freshness marts. Used incremental models for sanctions history, snapshots for SCD Type 2 license states, and dbt tests (not_null, unique, relationships, freshness) to guarantee NCQA-grade data quality. Orchestrated runs via Airflow with Airbyte-loaded sources.
CB1.5 Pipeline – Sanctions & License Ingestion	Lead Engineer & Architect	Designed and led CertifyOS's largest ingestion system, migrating 500+ manually processed license file types into a unified, job-config-driven ELT pipeline. Orchestrated end-to-end with Apache Airflow (Cloud Composer) , ingested sources via Airbyte , and applied dbt models in BigQuery. Achieved full automation of monthly state license refreshes, previously handled via separate scripts per file.
SCD Type 2 Utility	Designer & Developer	Built reusable dbt snapshot and incremental models on Snowflake and BigQuery for Slowly Changing Dimensions (Type 2) using hash-based change tracking. Applied across sanctions datasets lacking primary keys, ensuring accurate historical tracking and auditability with isCurrent, startDate, and endDate logic.
NPPES to DAL Transformation	Architect & Engineer	Developed dbt staging and mart models in BigQuery to convert flat nppes_raw tables into nested DAL-ready practitioner payloads for API ingestion. Coordinated with Platform team to validate schemas, manage chunked ingestion, and support 300+ column mappings.
Matching Engine for Sanctions Attribution	Co-designer	Co-developed a fuzzy matching engine to attribute provider sanctions to practitioners using name, address, and license matching. Critical for linking records lacking NPIs, improving attribution rate from 75% to 97%.
Scraper Automation Layer DAG	Builder & Coordinator	Created an Airflow DAG to orchestrate Airbyte syncs of scraper outputs into validation pipelines. Integrated with license enrichment and matching layers to support NCQA license freshness compliance. Worked closely with the Scraper team to finalize source contracts and ensure reliable ingestion.
NCQA License Refresh Pipeline	Designer	Enabled compliance by designing an Airflow-orchestrated system that ensures all licenses are refreshed within 30 days, powered by the scraper automation layer. Tracked freshness across license types and ensured seamless updates into internal compliance systems.
GitHub Actions Release Pipelines	DevOps Contributor	Implemented CI/CD workflows for Python package deployment to Google Artifact Registry with retry logic, version bumping, and concurrency controls. Segregated staging (release/stage-*) and production (release/main) workflows with semantic versioning and selective rebuilds.
CertifyOS API Mapping & Integration	Technical Lead	Mapped multiple flat datasets (NPPES, sanctions, licenses) to deeply nested API schemas. Ensured complete schema validation, null handling, type coercion, and custom field logic for payload ingestion into DAL endpoints.

Project	Role	Description
State Sanctions Dataflow Design (CB1.5)	Dataflow Architect	Designed the DataFlow logic for ingesting state sanctions files using a config-driven approach with schema and transformation logic maintained in BQ config tables. Enabled scalable file parsing and publishing of standardized records into Pub/Sub for downstream CB2.0 systems.

KEY IMPACT METRICS

Area	Impact
Provider Attribution	Improved from 75% to 97% using enriched matching engine
License Refresh	Migrated 500+ script-based processors to a single automated pipeline
Compliance (NCQA)	Ensured license data is refreshed within 30-day SLA
Data Quality	Enforced NCQA-grade integrity via dbt tests on Snowflake marts (freshness, uniqueness, referential checks)
Manual Effort	Reduced manual touchpoints to zero via config-driven pipeline design
Modeling Velocity	Reusable dbt macros and layered models accelerated new compliance dataset onboarding

EDUCATION

Bachelor of Engineering (Information Technology)

University of Pune, India – 2016

CAREER GOAL

To transition formally into an **Engineering Manager** or **Principal Data Architect** role — combining deep technical vision with team leadership to build scalable, resilient data platforms that drive critical business outcomes.

6sense

Senior Data Engineer

Oct 2022 – Sept 2024 | Bengaluru, India (Remote/Hybrid)

SUMMARY

At 6sense, a leading AI-driven Account-Based Marketing (ABM) and Revenue Intelligence platform, I served as a Senior Data Engineer within the Big Data Platform team. I helped design and implement scalable internal tools and frameworks used across 6sense’s real-time and batch data systems — powering customer-facing

insights, lead enrichment, and campaign optimization. I led the rollout of a **dbt-on-Snowflake** analytics layer, migrating brittle hand-written SQL into tested, version-controlled dbt models and standing up Snowflake as the curated reporting warehouse for Marketing Ops, SalesOps, and Data Science. I also contributed to platform-wide initiatives supporting one of the **largest big data clusters in the industry**, with over **1 million jobs processed daily** across Spark, Trino, and Delta Lake pipelines.

KEY PROJECTS

Project	Role	Description
Snowflake Analytics Warehouse Migration	Lead Developer	Stood up Snowflake as the curated analytics warehouse for ABM and revenue-intelligence reporting. Designed multi-layer schemas (raw → staging → marts), warehouse sizing, RBAC, and resource monitors — cutting query latency and giving analysts self-serve access independent of Spark clusters.
dbt Transformation Framework	Lead Developer	Built 6sense’s dbt project on Snowflake: 120+ models with staging/intermediate/mart layering, incremental models for high-volume lead-enrichment tables, snapshots for SCD tracking, plus schema/freshness tests and exposures wired into CI. Eliminated duplicated SQL and made transformations testable and documented.
Singlestore ETL Framework	Lead Developer	Led development of a unified ETL framework for managing ingestion, transformation, and archival of real-time and batch data via Airbyte into Singlestore and Snowflake , with Trino-Hive integration, auto-scaling, and cross-region deployments feeding the dbt transformation layer.
Data Extractor Utility	End-to-End Owner	Configured Airbyte source connectors for extracting and archiving data from MySQL, Postgres, and Singlestore into S3/HDFS with configurable filtering, partitioning, and retention rules without code changes. Streamlined compliance and ops workflows.
Hive Table Cloning Utility	Lead Developer	Created a utility to rapidly clone production Hive tables to dev environments for staging/debugging. Reduced manual time from hours to minutes with built-in error handling and flexible config support.
Delta & Spark Batch Job Optimization	Contributor	Improved performance of legacy Spark jobs handling TB-scale lead enrichment data. Reduced job run time and optimized partitioning and shuffle logic across multiple data domains.
Cross-Team Data Contract Ownership	Collaborator	Worked with Product, Marketing, and Analytics teams to manage data schema consistency, ingestion failures, and format evolution — reducing downstream breakages and improving SLAs.

KEY IMPACT METRICS

Area	Impact
Analytics Self-Serve	Delivered 120+ tested dbt models on Snowflake, removing engineering bottlenecks for reporting

Area	Impact
Warehouse Cost	Tuned Snowflake warehouse sizing and auto-suspend to control compute spend
Data Quality	dbt tests caught schema/freshness regressions before they reached dashboards
Platform Efficiency	Reduced ingestion and job latency across Spark & Delta jobs by tuning partitioning and I/O
Developer Productivity	Enabled rapid table cloning and pipeline debugging across staging environments
Data Consistency	Standardized schema contract validation across cross-functional teams
Cost Optimization	Participated in AWS S3/HDFS archival strategy to reduce storage and query costs
ETL Framework Adoption	Internal tooling adopted org-wide for managing batch and CDC ingestion pipelines

TECH STACK

- **Modern Data Stack:** Apache Airflow, dbt, Airbyte, Snowflake
- **Languages:** Python, PySpark, SQL
- **Compute & Storage:** Singlestore, Hive, Trino/Presto, Delta Lake, Spark, HDFS, S3
- **Databases:** MySQL, PostgreSQL, RDS
- **Streaming & Messaging:** Apache Kafka
- **Cloud Infrastructure:** AWS (S3, Lambda, RDS, Kinesis)
- **Orchestration & Monitoring:** Airflow, custom internal schedulers, Spark job orchestration

COLLABORATION & RESPONSIBILITIES

- Served as the **Lead Developer** on key internal utilities used across the Big Data team
- Collaborated with Staff Engineers and Platform Leads on **multi-region replication, CDC strategies, and performance benchmarking**
- Worked cross-functionally with **Marketing Ops, SalesOps, and Data Science** to meet SLA, contract, and delivery needs
- Participated in monitoring and maintaining high-throughput Spark and Trino jobs that support 6sense’s ABM intelligence layers
- Supported platform-wide initiatives to reduce AWS resource usage and improve system resilience

InCred Financial Services Ltd

Tech Lead – Data Engineer

Apr 2021 – Sept 2022 | Bengaluru, India

SUMMARY

At InCred, a digital-first lending and financial services provider, I served as the **Tech Lead for the Data Platform team**, driving the architecture and development of a robust, real-time data ecosystem that powered analytics, credit risk modeling, and reporting across the business. I owned end-to-end data platform initiatives — from ingestion to lakehouse modeling — while mentoring engineers and analysts, optimizing infrastructure, and establishing data reliability and cost efficiency as core principles.

KEY PROJECT

Project	Role	Description
InCred Data Platform	Tech Lead & Platform Architect	Designed and led the development of a near real-time data platform ingesting CDC data from MySQL, Postgres, and DynamoDB using Airbyte , and transforming it in Databricks using PySpark, Autoloaders, and dbt models . Delivered clean, queryable Delta Lakes in S3 and Snowflake marts to power analytics, ML pipelines, and dashboards. Led cost optimization efforts that reduced infra spend by 30% , mentored a team of 4, and implemented monitoring, retry, and resource-tuned Databricks pools to minimize job latency.

KEY IMPACT METRICS

Area	Impact
Infrastructure Cost	Reduced streaming + ETL cost by over 30% through optimized pipeline design
ETL Latency	Cut job start time and report refresh delays by 30% with Databricks pool tuning
System Reliability	Migrated unstable on-prem Kafka-Dynamo pipelines to AWS-native Kinesis Firehose , improving uptime and observability
Team Growth	Mentored and upskilled a team of 4+ engineers/analysts across PySpark, streaming, and data contract practices

TECH STACK

- **Modern Data Stack:** Apache Airflow, dbt, Airbyte, Snowflake
- **Languages & Processing:** Python, PySpark, Spark Streaming
- **Data Engineering Platforms:** Databricks (Jobs, Autoloader, MLFlow), AWS DMS
- **Streaming Systems:** AWS Kinesis, Kafka, SQS
- **Storage & Lakehouse:** Delta Lake on S3, EFS
- **Databases:** MySQL, PostgreSQL, DynamoDB
- **Monitoring & Scheduling:** Airflow, Databricks jobs, alerting, retry logic
- **Analytics:** Presto, Metabase
- **ML Collaboration:** Supported feature stores in Druid; integrated with MLFlow model tracking pipelines

LEADERSHIP & COLLABORATION

- Acted as **Tech Lead and Platform Owner** for the company-wide data platform
- Solely responsible for **architecting CDC ingestion workflows** and tuning streaming pipelines for stability and cost efficiency
- Partnered with ML teams to build **feature stores** and ensure high-frequency data availability for model training
- Enabled Risk, Analytics, and Product teams to access **reliable, fresh, and performant data**
- Mentored junior engineers in **streaming design patterns, Databricks optimization, and infrastructure monitoring**

Slice

Data Engineering Lead → Data Engineer

Nov 2019 – Apr 2021 | Bengaluru, India

SUMMARY

At Slice, a fast-scaling consumer credit and payments startup, I transitioned from an individual contributor to **Data Engineering Lead**, building the foundation of the company's data infrastructure, warehouse, and tooling. I architected core systems like the **Data Lake, User Graph, and data propagators**, while also mentoring the growing data team and owning stakeholder relationships across risk, analytics, and product teams. My contributions helped **de-risk infra scale-up, enable self-serve analytics, and cut costs** on third-party tools by driving internal platform innovation.

KEY PROJECTS

Project	Role	Description
Slice Data Lake	Lead Developer	Architected and implemented a PySpark + S3-based data lake that offloaded analytical workloads from MongoDB to Delta-compatible Parquet storage on AWS. Enabled real-time ingestion using Kinesis, Glue/EMR, and Airbyte , freeing MongoDB for high-throughput API workloads and reducing infra costs by ~\$10K/month.
Slice User Graph	Lead Developer	Built a graph-based analytics engine using AWS Neptune to identify clusters of users with shared risk/revenue traits. Powered user approval/rejection logic by analyzing links to "good" or "risky" users. Improved underwriting strategies and fraud detection accuracy.
S3 to Snowflake/RDS Propagator	Lead Developer	Developed a user-friendly tool for analysts and PMs to safely push data to Snowflake and RDS without needing engineering intervention. Included validation, schema mapping, and write protections to avoid data corruption.
Mongo to Snowflake Pipeline	Lead Developer	Built an Airbyte CDC pipeline for transactional and behavioral user data from MongoDB to Snowflake , with dbt models for analytics marts. Enabled deeper analysis by the Risk team and eliminated dependency on expensive third-party tools (~\$600/month saved).

Project	Role	Description
Jarvis: Slack Notification Framework	Lead Developer	Created an internal Slack-based alerting system triggered by SNS, S3 events , and job outcomes. Widely adopted across teams for real-time ops visibility and job health monitoring. Plug-and-play for any Slack channel.

KEY IMPACT METRICS

Area	Impact
Infrastructure Cost	Saved ~\$10,000/month by offloading MongoDB and ~\$600/month from third-party tool replacement
Platform Reliability	Improved real-time ingestion stability and freed MongoDB for API performance
Analytics Enablement	Empowered analysts and PMs to push/report data without engineering dependency
Fraud Risk Mitigation	Enabled graph-based user linking for approval/rejection logic
Team Growth	Built and led the data engineering team from scratch, mentoring new engineers on AWS and Spark tech stack

TECH STACK

- **Modern Data Stack:** Apache Airflow, dbt, Airbyte, Snowflake
- **Languages:** Python, PySpark
- **Data Infra:** AWS S3, Glue, EMR, Kinesis, Athena, Amazon Redshift, Snowflake, RDS, EC2, DynamoDB
- **Databases:** MongoDB, AWS Neptune
- **Eventing & Messaging:** AWS SNS, S3 Events, SQS
- **Alerting & Observability:** Slack APIs, Custom Slack Bot Framework (Jarvis)

LEADERSHIP & COLLABORATION

- **Promoted** from IC to **Data Engineering Lead** within 12 months based on technical ownership and delivery
- Led a **team of 3 engineers** while working directly with analytics, product, and risk teams
- Designed team processes and mentored junior members on AWS tools, infrastructure best practices, and PySpark
- Played a critical role in **selecting and deploying foundational data systems**, balancing cost, speed, and reliability
- Drove technical blogs to document internal tooling (e.g., **Jarvis, Data Propagator, User Graph**)

Particle41 India LLP

Software Developer

SUMMARY

At Particle41, my first role post-college, I progressed from an individual contributor to a **lead developer and systems architect** across multiple domains — from **big data processing and identity resolution** to **ML pipelines and full-stack applications**. I collaborated directly with international clients, including U.S.-based CTOs, to deliver production-grade solutions. My hands-on work across **AWS and GCP**, combined with **cross-functional ownership**, helped establish a strong technical foundation that continues to support my engineering leadership today.

KEY PROJECTS

Project	Role	Description
Onboarding Engine (IDify)	Lead Developer & Architect	Designed and implemented a scalable Spark-based ETL pipeline on AWS EMR , processing CRM data from partner clients into digital user profiles. The system included a matching engine, tag server, and reporting layer, enabling high-throughput identity onboarding and significantly enhancing online audience match rates.
Identity Resolution (ID Graph)	Lead Developer & Architect	Developed a user identity graph using GraphFrame on Spark to connect identifiers like cookies, devices, and emails into unified user profiles. Improved downstream personalization and attribution by enhancing entity linkage.
DFP ROI Enhancements (Vevo)	Team Lead	Led a team to build a rule-generation ML pipeline using PySpark + MLLib , improving Vevo’s ad targeting performance and on-target impression ROI.
Program Management Report	Developer	Delivered an ETL system to ingest third-party buyer reports and generate automated revenue dashboards using Python (PETL) and Redshift . Also gathered client requirements firsthand.
MiMedia FTE (Android)	Developer	Enhanced the user onboarding experience in the Android app using animated FTE screens, improving engagement and retention.
Online Catalog Viewer (RepSpark)	Lead Developer	Led the development of a React + C# web app to help wholesalers build interactive digital product catalogs and take B2B orders in real time.

KEY IMPACT METRICS

Area	Impact
Data Product Scale	Delivered Spark- and EMR-based pipelines that scaled with onboarding growth
Client Satisfaction	Worked directly with the CTO of ALC (a U.S.-based client), who credited me as “a critical part of our success” due to my reliability, technical depth, and leadership in onboarding and incident resolution

Area	Impact
Team Enablement	Trained new engineers and led delivery across multiple team transitions
System Design	Progressed from developer to platform architect across multiple distributed systems
Cost Optimization	Replaced manual workflows and third-party tooling with custom-built, efficient solutions

TECH STACK

- **Big Data & Processing:** Apache Spark, Hive, GraphFrame, PySpark
- **Cloud Platforms:** AWS EMR, S3, EC2, Glue, Lambda, Athena, GCP (limited scope)
- **Languages & Tools:** Python 2.7/3, PETL, Java (Android), C#.NET, ReactJS
- **Databases:** PostgreSQL, DynamoDB, Microsoft SQL Server, Redshift
- **ML & Analytics:** Spark MLlib
- **Dev & Deployment:** Slack APIs, CI pipelines, documentation ownership

RESPONSIBILITIES & GROWTH

- Transitioned from graduate hire to **lead developer and system designer** within 2.5 years
- Mentored peers and junior engineers, particularly during team transitions and onboarding
- Directly engaged with global clients (U.S. and EU), understanding requirements and delivering production-quality solutions
- Received praise from **Rick Landsman (CTO of ALC)** for playing a crucial role in the success of mission-critical systems and for consistently leading incident resolution, daily team syncs, and onboarding
- Owned technical delivery across identity resolution, onboarding engines, analytics reporting, and full-stack UI projects

Lifelong Learning & Foundations

CERTIFICATIONS

- **Introduction to Big Data** – Coursera | Oct 2019 [View Certificate](#) *Explored big data architecture, processing frameworks like Hadoop and Spark, and distributed data pipelines.*
- **SQL Fundamentals** – SoloLearn | May 2016 [View Certificate](#) *Built core skills in SQL, data querying, joins, aggregations, and schema operations.*

PUBLICATIONS

- **Stock Market Forecasting Using Hybrid Methodology** – *Imperial Journal of Interdisciplinary Research (IJIR)*, May 2016 [Read Publication](#) *Co-authored a research paper on forecasting stock trends using statistical and machine learning models.*
-

EARLY CAREER FOUNDATION

- **Project Intern – Persistent Systems, Pune** | Sept 2015 – July 2016 *Worked on a sponsored academic project titled “Stock Market Prediction using Technical Analysis.” Developed the system using Django, HTML, CSS, and JavaScript. Gained firsthand experience in project scoping, timelines, and real-world workplace culture.*

Education

Degree	Institution	Year	Grade
Bachelor of Engineering (IT)	Pune University	July 2016	73.08%
Senior Secondary (12th)	Kendriya Vidyalaya No. 3, 9BRD, Pune	2010	82.20%
Secondary (10th)	Kendriya Vidyalaya No. 3, 9BRD, Pune	2008	91.20%

Key Achievements & Technical Competitions

- **ACM ICPC Amritapuri Regional Onsite Contestant** – 2014
- **1st Prize** – C Programming Contest (SE), Sinhgad Academy of Engineering, Pune (2013)
- **2nd Prize** – CodeChef Challenge, AISSMS College Fest (2014)
- **3rd Prize** – C-Athlon, Melange Fest, BRACT's VIT, Pune (2014)
- **Finalist** – C-Maestros, MITCOE Tesla 2014
- **Finalist** – Oracle Think.com Web Page Contest (2007–08)
- **Qualified** – 1st Rounds of **Facebook HackerCup** and **Google CodeJam**, 2015
- **Global Rank #242** – TCS CodeVita 2014
- **India Rank #619** – HackerRank CodeSprint India 2014
- **Waitlisted Finalist** – CSI National Programming Contest 2014
- **Runner-Up** – Chess Tournament, INNOFEST 2013, Sinhgad Academy of Engineering
- **U-19 Cricket Player** – Represented Kendriya Vidyalaya Sangathan, Mumbai Region (2011–12)

Leadership & Volunteer Experience

- **HackerRank Campus Ambassador** and *Founder of Coders Club* on campus
- **Student Council Member** – 2014–15 academic year
- **Head Student Coordinator** – *Freaky 'C'oders*, TECHTONIC 2015
- **Head Event Coordinator** – *Freaky 'C'oders*, TECHTONIC 2016